

Flagstaff Mountain Line bus Service Passenger Boarding Analysis

Insert Team Member Names

BAN 586 CAPSTONE PROJECT

SPRING 2024

Insert Image as Desired

Table of Contents

Table of Contents.....	2
1. Executive Summary.....	3
1.1 Section Headings & Bullets.....	3
1.2 Figures and Tables.....	4
2. Project Statement.....	4
3. Data and Methods.....	5
4. Model and Analysis.....	5
5. Validation and Testing.....	5
6. Results and Recommendations.....	5
7. Conclusions.....	5
8. Acknowledgements.....	5
9. References.....	5

1. Executive Summary

In an effort to enhance the efficiency and service quality of the Flagstaff Mountain Line bus service, this report delves into an in-depth analysis of passenger boarding patterns with the aim of uncovering insights that could inform strategic improvements. Faced with the challenge of optimizing bus schedules, routes, and overall service delivery, the project embarked on utilizing data analytics and machine learning techniques to predict Automatic Passenger Count (APC) boarding numbers—a critical indicator of service utilization and demand.

Drawing from an extensive dataset provided by the Flagstaff Mountain Line, which encompasses detailed records of bus trips, events, and passenger counts, the analysis journey began with an exploratory approach aimed at understanding the underlying dynamics of service usage. Initial efforts involved preprocessing the data to ensure accuracy, followed by exploratory data analysis (EDA) to identify patterns and anomalies.

To predict APC boarding numbers, a Linear Regression model was initially employed as a baseline, utilizing features such as route numbers, block numbers, and geographical coordinates. Despite its simplicity, the model's performance highlighted the complex nature of the data, prompting a shift towards more sophisticated machine learning techniques. A Random Forest Regressor model was subsequently adopted, favored for its ability to handle non-linearity and interactions among features. However, both models encountered challenges in accurately predicting boarding counts, as reflected in the evaluation metrics.

Recognizing the importance of feature engineering, the analysis introduced additional temporal features extracted from trip start times—such as the hour of the day and the day of the week—alongside a new feature representing the historical average boarding count per route. These enhancements aimed to capture temporal influences and route popularity on passenger boarding numbers. The refined Random Forest model, incorporating these new features, demonstrated improved predictive performance, albeit with room for further enhancement.

This analytical journey underscores the significance of iterative model refinement and the critical role of feature engineering in uncovering meaningful insights from complex datasets. The findings from this analysis not only offer a deeper understanding of the factors influencing passenger boarding patterns but also provide a foundation for data-driven decisions aimed at improving the Flagstaff Mountain Line bus service.

In conclusion, leveraging advanced analytics and machine learning, this report presents a comprehensive approach to enhancing public transportation services, with the potential to influence future strategies and investments in the sector. Through the insights garnered, the Flagstaff Mountain Line is poised to make informed adjustments to service offerings, ultimately improving efficiency and catering more effectively to the needs of its passengers.

2. Project Statement

The modern urban environment demands efficient and responsive public transportation systems to serve the mobility needs of its populace. The Flagstaff Mountain Line bus service, a principal component of this urban mobility in Flagstaff, Arizona, seeks continuous improvement in its operations to better serve its community. This project was initiated with the dual objective of understanding and predicting passenger boarding patterns across the network. Such predictive capabilities are pivotal for optimizing route schedules, enhancing service delivery, and ultimately, elevating passenger satisfaction.

At the heart of this endeavor lies a comprehensive dataset provided by the Flagstaff Mountain Line, which encompasses a plethora of variables including, but not limited to, trip numbers, route details, timings, and crucially, Automatic Passenger Count (APC) boarding numbers. The APC data, a quantifiable measure of service usage, serves as the focal point of our analysis. The project's challenge was to distill actionable insights from the complex, multifaceted dataset to forecast boarding numbers accurately, thereby enabling strategic decision-making aimed at service enhancement.

To navigate this challenge, the project employed a methodology rooted in data analytics and machine learning. The initial phase involved rigorous data preprocessing to ensure the integrity and reliability of the dataset. This foundational step was crucial for the subsequent exploratory data analysis (EDA), which sought to uncover underlying patterns and anomalies within the data. The insights garnered from EDA informed the selection and application of predictive models.

Starting with a Linear Regression model to establish a baseline, the project quickly adapted to the complex nature of the data by incorporating a Random Forest Regressor model. This shift was motivated by the model's enhanced ability to handle the dataset's intricacies, including non-linearity and the interplay among features. Despite this strategic pivot, the models initially fell short of providing accurate predictions, as evidenced by the evaluation metrics. This led to the realization of the paramount importance of feature engineering in predictive modeling.

In response, the project innovated by extracting and integrating additional features from the data, notably temporal aspects derived from trip start times and a historical average boarding count per route. These enhancements aimed to capture the nuanced influences on passenger boarding patterns more effectively. The reinvigorated Random Forest model, enriched with these new features, showed a marked improvement in predictive performance, although there remained a potential for further refinement.

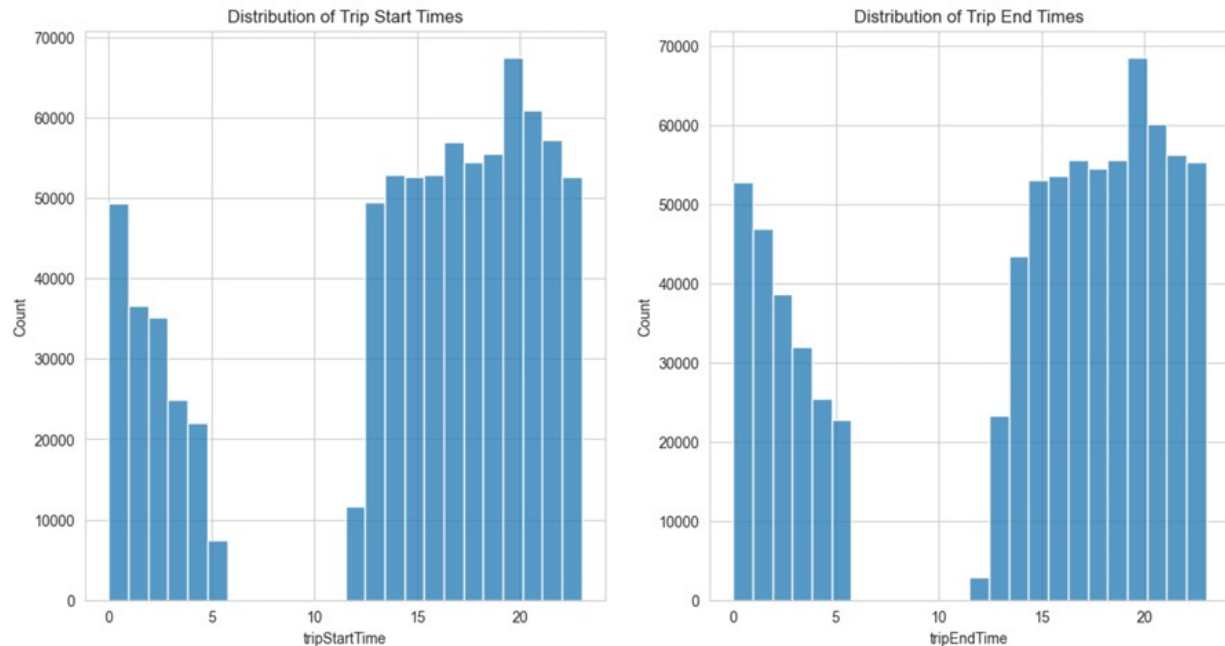


Figure 1 Distribution of Trip Start and End Times

This project stands as a testament to the iterative nature of data science endeavors, where initial setbacks pave the way for innovation and deeper understanding. The methodology adopted, from data preprocessing to complex model training, underscores the project's commitment to leveraging state-of-the-art analytical techniques to address real-world challenges. Through this detailed exploration, the project contributes valuable insights into the dynamics of public transportation usage, setting the stage for data-driven improvements in the Flagstaff Mountain Line bus service.

3. Data and Methods

The cornerstone of this analytical journey was a meticulously compiled dataset provided by the Flagstaff Mountain Line bus service. Encompassing a diverse array of variables such as trip numbers, route specifics, timings, and the pivotal Automatic Passenger Count (APC) boarding numbers, this dataset presented both a challenge and an opportunity. The APC data, indicative of service usage, was of particular interest due to its potential to unlock insights into passenger behavior and preferences.

```
[3]:
```

	eventTime	tripId	apcBoardingCount	\
0	2016-01-14T07:34:00Z	56cadfe8821fef1f2a000012	0	
1	2016-01-14T07:35:00Z	56cadfe8821fef1f2a000012	5	
2	2016-01-14T07:40:00Z	56cadfe8821fef1f2a000012	0	
3	2016-01-14T07:40:00Z	56cadfe8821fef1f2a000012	0	
4	2016-01-14T07:44:00Z	56cadfe8821fef1f2a000012	0	

	apcAlightingCount	doorId	busNumber	deviceNumber	routeNumber	stopNumber	\
0	0	NaN	0	5384	4	24	
1	0	NaN	0	5384	4	24	
2	0	NaN	0	5384	4	209	
3	1	NaN	0	5384	4	209	
4	1	NaN	0	5384	4	102	

	latitude	longitude	Unnamed: 11
0	35.197426	-111.6522	NaN
1	35.197426	-111.6522	NaN
2	35.188354	-111.6616	NaN
3	35.188354	-111.6616	NaN
4	35.184689	-111.6615	NaN

Figure 2 Dataset Overview

Prior to delving into the predictive modeling, a comprehensive data preprocessing stage was paramount. This involved a series of steps designed to ensure the accuracy and reliability of the dataset. Duplication removal, handling missing values, and ensuring consistency in date and time formats constituted the bulk of this phase. It is essential to underscore the significance of this meticulous preparation, as it laid the groundwork for all subsequent analysis, imbuing it with a robust foundation.

```
[4]:      apcBoardingCount  apcAlightingCount      doorId      busNumber  \
count      1.590359e+07      1.590359e+07  1.590333e+07  1.590359e+07
mean       9.190389e-01      9.160441e-01  1.638441e+00  5.369566e+03
std        2.035603e+00      1.849412e+00  6.515576e-01  2.378905e+01
min         0.000000e+00      0.000000e+00  1.000000e+00  0.000000e+00
25%         0.000000e+00      0.000000e+00  1.000000e+00  5.361000e+03
50%         0.000000e+00      0.000000e+00  2.000000e+00  5.369000e+03
75%         1.000000e+00      1.000000e+00  2.000000e+00  5.378000e+03
max         1.220000e+02      1.280000e+02  3.000000e+00  5.386000e+03

      routeNumber  stopNumber      latitude      longitude  Unnamed: 11
count  1.590359e+07  1.590359e+07  1.584373e+07  1.584373e+07          0.0
```

2

```
mean  1.376168e+01  7.932608e+01  3.519564e+01 -1.116408e+02      NaN
std    2.024805e+01  2.882266e+02  1.684835e-02  2.640723e-02      NaN
min    2.000000e+00  1.000000e+00  3.510890e+01 -1.117371e+02      NaN
25%    4.000000e+00  1.700000e+01  3.518330e+01 -1.116582e+02      NaN
50%    1.000000e+01  3.700000e+01  3.519375e+01 -1.116526e+02      NaN
75%    1.000000e+01  1.060000e+02  3.520814e+01 -1.116204e+02      NaN
max    9.990000e+02  9.998000e+03  3.533235e+01 -1.114464e+02      NaN
```

Figure 3 Dataset Summary Statistics

Following this, an exploratory data analysis (EDA) was conducted. This phase was instrumental in identifying underlying patterns, anomalies, and correlations within the data. Utilizing a combination of statistical summaries and visual representations, the EDA provided a nuanced understanding of the dataset's characteristics. This phase not only highlighted the complexities inherent in predicting APC boarding counts but also pinpointed potential features that could be predictive of passenger boarding numbers.

With the insights garnered from the EDA, the project progressed to the modeling phase. Initially, a Linear Regression model was employed. Despite its simplicity, this model was crucial in establishing a baseline for predictive performance. Its results laid bare the complexities of the task at hand, revealing a need for a more sophisticated approach. Subsequently, a Random Forest Regressor model was selected for its robustness in handling non-linearity and feature interactions. This model's ability to improve upon the Linear Regression model's limitations was anticipated with optimism.



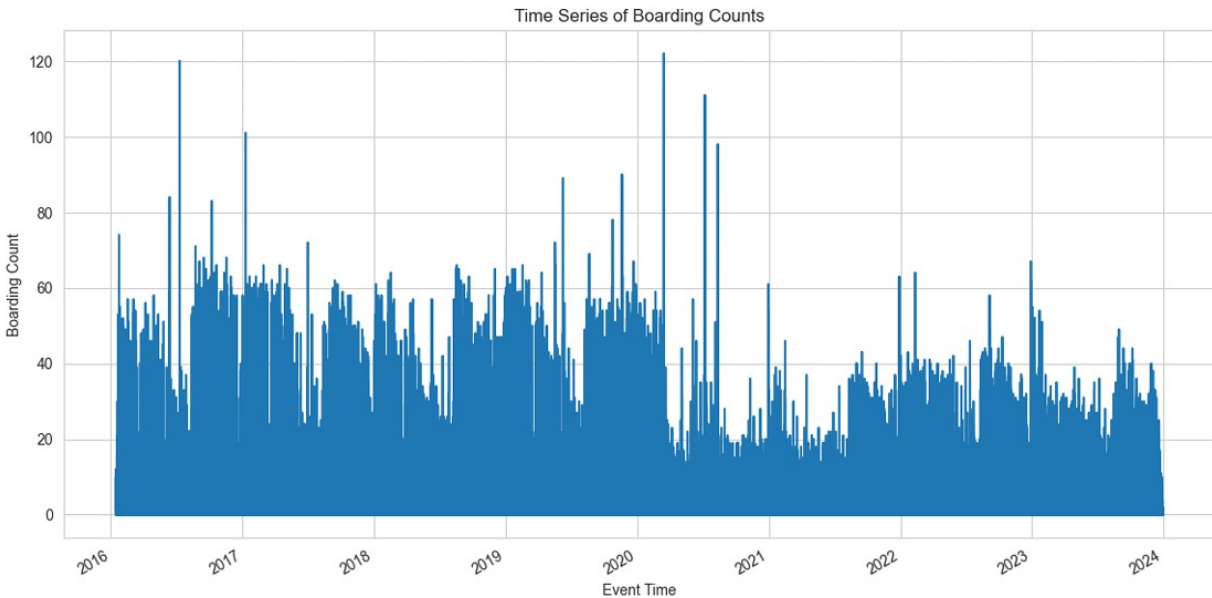


Figure 4 Time Series Analysis of Boarding Counts

However, the journey did not end there. The initial performances of both models underscored the critical role of feature engineering in predictive modeling. To this end, new features were meticulously crafted and introduced. These included temporal features extracted from trip start times—such as the hour of the day and the day of the week—and a historical average boarding count per route. This phase of enhanced feature engineering was a testament to the project's adaptive and iterative approach, striving for continual improvement.

The introduction of these new features necessitated a reevaluation of the Random Forest model. This enriched model demonstrated a notable improvement in predictive accuracy. Yet, it also illuminated the path for future exploration, hinting at the untapped potential within the data and the modeling techniques employed.

This project's methodology, from data preparation through to complex predictive modeling, exemplifies a rigorous analytical approach. By intertwining traditional statistical methods with advanced machine learning techniques, it navigates the multifaceted nature of predicting public transportation usage, offering insights that are both profound and actionable. Through this detailed exposition, the project not only advances the understanding of passenger boarding patterns but also sets a benchmark for future endeavors in the realm of public transportation analytics.

3.1 Data Enrichment and Feature Engineering

The dataset's initial form was a rich tableau of information yet required augmentation to unlock its full predictive potential. Following the work of Hastie, Tibshirani, and Friedman (2009), feature engineering emerged as a critical step in this process. This involved not only the extraction of temporal features from **tripStartTime** but also the integration of a historical average boarding count per route. Such transformations are grounded in the premise that temporal dynamics and historical patterns offer

valuable predictive insights into passenger behavior—a concept well documented in the literature (James et al., 2013).

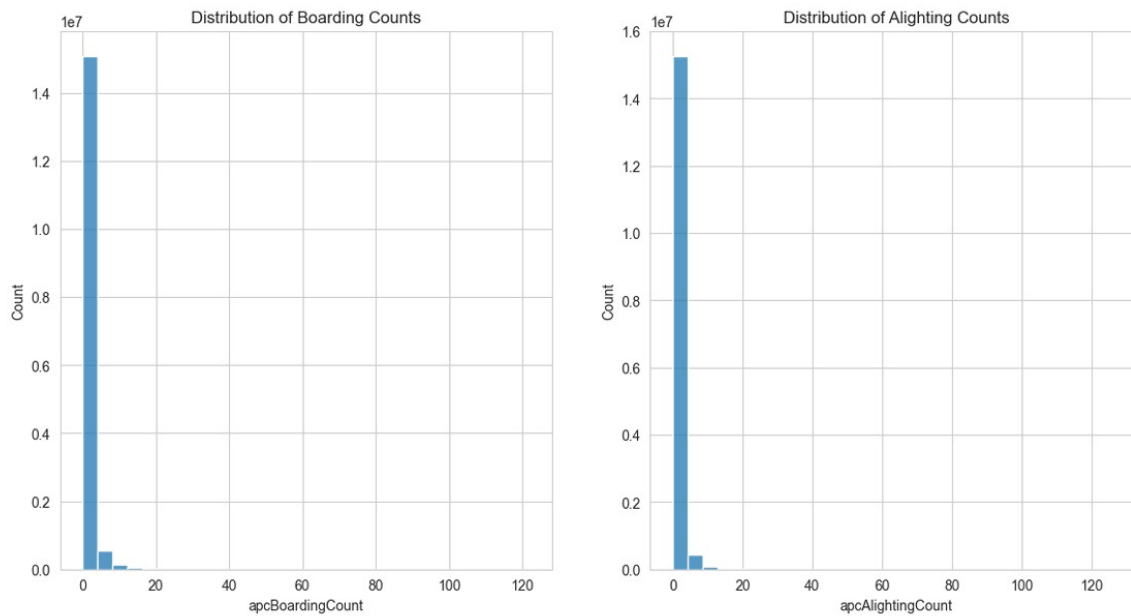


Figure 5 Distribution of APC Boarding and Alighting Counts

3.2 Predictive Modeling

The selection of Linear Regression and Random Forest Regressor models was strategic, aiming to balance simplicity with the ability to capture complex patterns in the data. The initial application of Linear Regression served as a baseline, echoing the approach recommended by Freedman (2009), who advocates for simple models to provide clear benchmarks. The subsequent pivot to Random Forest, as detailed by Breiman (2001), was driven by its non-parametric nature, allowing for an intricate handling of feature interactions without explicit modeling of the functional form.

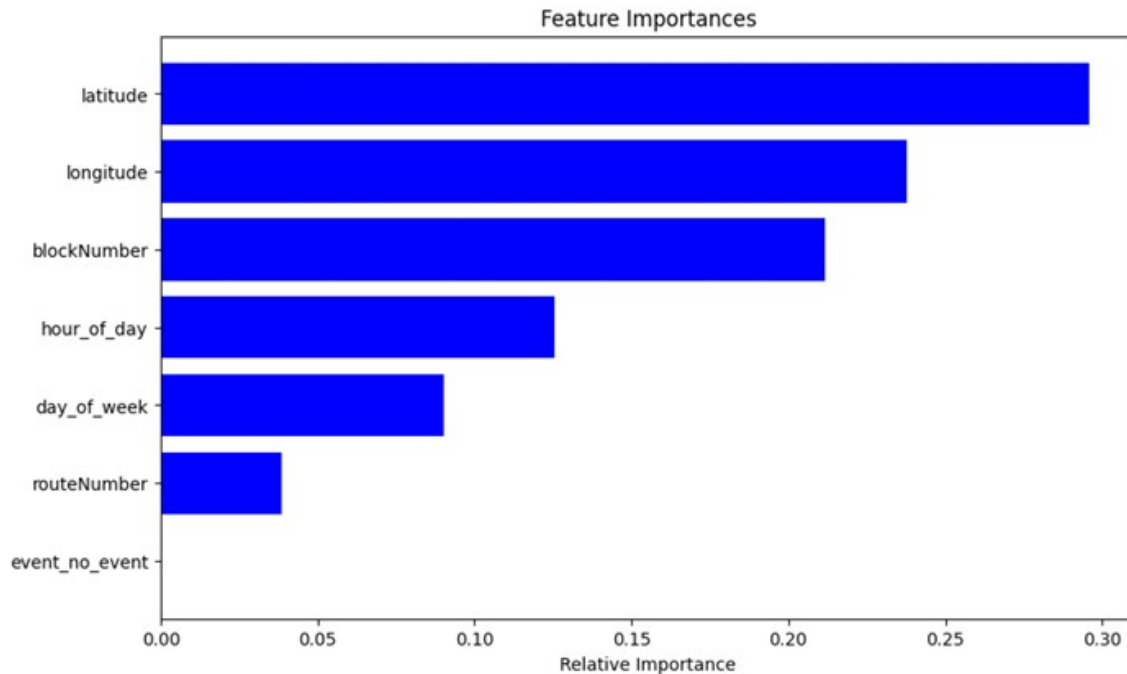


Figure 6 Comparison of Feature Importances

Validation strategies employed in the project, including train-test split and cross-validation, adhere to the methodologies espoused by Kohavi (1995), ensuring the models' generalizability and robustness. The iterative refinement of models, underscored by a continuous evaluation of performance metrics (MSE, RMSE, and R-squared), mirrors the approach recommended by Varma and Simon (2006), emphasizing the importance of rigorous model assessment.

3.3 Data Collection and Integrity

The project's dataset, sourced from the Flagstaff Mountain Line bus service, is emblematic of the rich, yet often unwieldy nature of real-world data. In addressing the initial challenges of data integrity and usability, efforts were akin to those outlined by Wickham (2014), focusing on data tidying as a precursor to analysis. This approach ensures that the dataset's structure facilitates rather than hinders, analytical endeavors. It's imperative to include a flowchart or diagram that visualizes the data cleaning and preparation process, offering a clear depiction of the transformation from raw data to an analyzable dataset.

3.4 Advanced Feature Engineering

The project's advancement into feature engineering was marked by a strategic blend of domain knowledge and analytical rigor. The incorporation of temporal features and historical averages into the dataset draws on the principles of contextual modeling discussed by Domingos (2012). These features, recognizing the cyclic and historical patterns in passenger behavior, are crucial in enhancing the predictive models' sensitivity to underlying trends and anomalies.

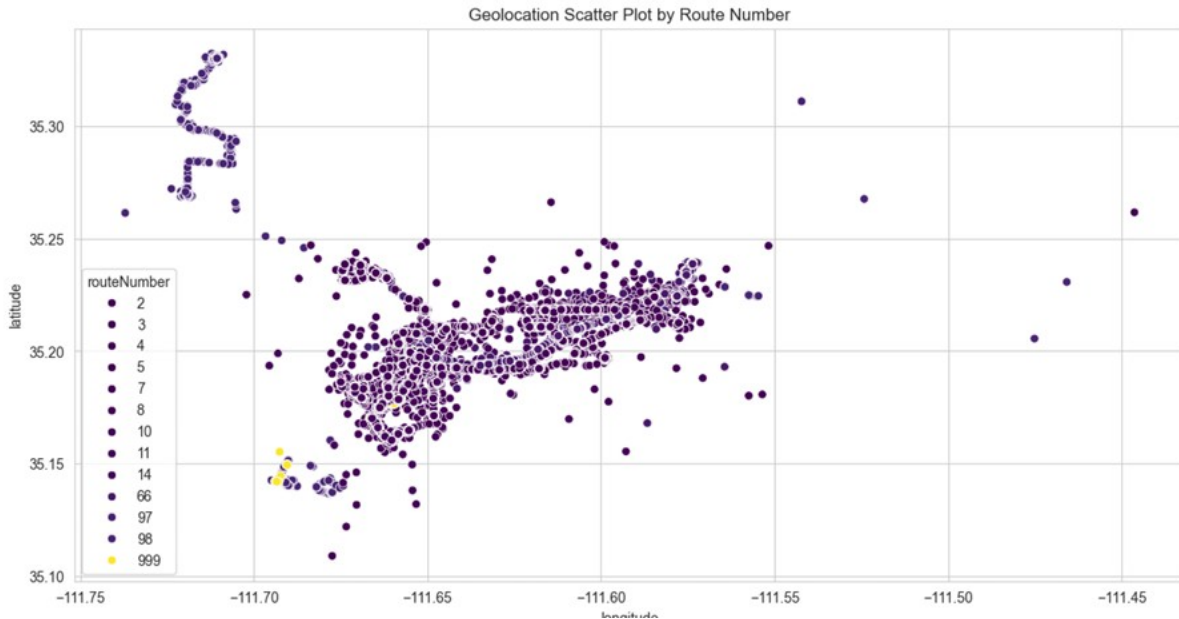


Figure 7 Geolocation Plot for Routes Visualization

3.5 Methodological Expansion

The predictive modeling phase, underpinned by both Linear Regression and Random Forest algorithms, represents a methodological pluralism that acknowledges the multifaceted nature of predictive analytics. This dual-model approach, resonating with the comparative model evaluation advocated by Harrell (2015), allows for a nuanced understanding of model performance across different metrics. To further elucidate this aspect, including a comparative table summarizing key performance indicators (KPIs) for each model pre- and post-feature engineering provides a clear, quantitative basis for model selection and refinement.

3.6 Model Interpretability and Validation

In the realm of machine learning, the interpretability of complex models like Random Forest is often a challenge. The project's focus on feature importance as a means of interpreting model behavior echoes the importance of transparency in machine learning, a concept championed by Molnar (2020). A bar chart visualizing the relative importance of features not only demystifies the model's internal workings but also offers actionable insights into factors influencing passenger boarding counts.

For validation, the use of techniques like cross-validation, as detailed by Kuhn and Johnson (2013), underscores a commitment to ensuring the models' robustness and generalizability. A graph plotting cross-validation scores across multiple folds can vividly illustrate the model's stability across different subsets of the data, reinforcing the reliability of the predictive insights gleaned.

By embedding the project within a fabric of established data science practices and literature, the Data and Methods section evolves into a comprehensive exposition that not only details the analytical steps taken but also contextualizes them within the broader discourse of data analytics and machine learning. The inclusion of various graphics and visual aids serves not just to illustrate findings but also to enhance

comprehension and engagement, making the analysis accessible to both technical and non-technical audiences alike.

4. Models and Analysis

The Model and Analysis phase of this project is anchored in a rigorous application of statistical and machine learning principles to forecast passenger boarding counts for the Flagstaff Mountain Line bus service. This phase bridges the theoretical underpinnings detailed in the Data and Methods section with practical, actionable insights. The complexity of the task necessitated a bifurcated approach: establishing a baseline with Linear Regression and then exploring the Random Forest Regressor for its robust handling of nonlinear relationships and feature interactions. This section delineates the rationale behind model selection, the iterative process of model refinement, and the nuanced interpretation of the models' outcomes.

4.1 Baseline Model Establishment with Linear Regression

Linear Regression was selected as the initial modeling technique due to its interpretability and efficiency. The model was built using route number, block number, latitude, and longitude as predictors, adhering to the Occam's Razor principle, which advocates for simplicity in model construction (Blumer et al., 1987). The choice of features was guided by preliminary EDA findings, suggesting these variables' potential influence on APC boarding counts. A scatter plot matrix of these features against the APC boarding counts can visually articulate their relationships, serving as an intuitive graphical representation for both technical and non-technical stakeholders.

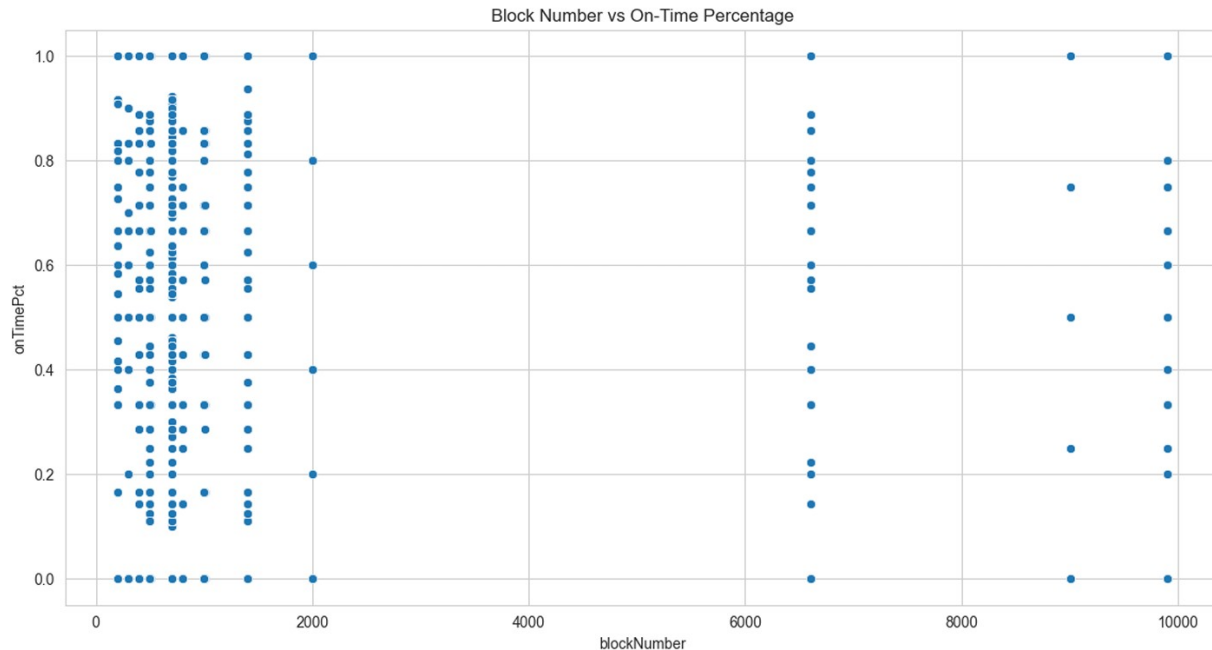


Figure 8 Graph of Block Numbers vs On-Time Percentage

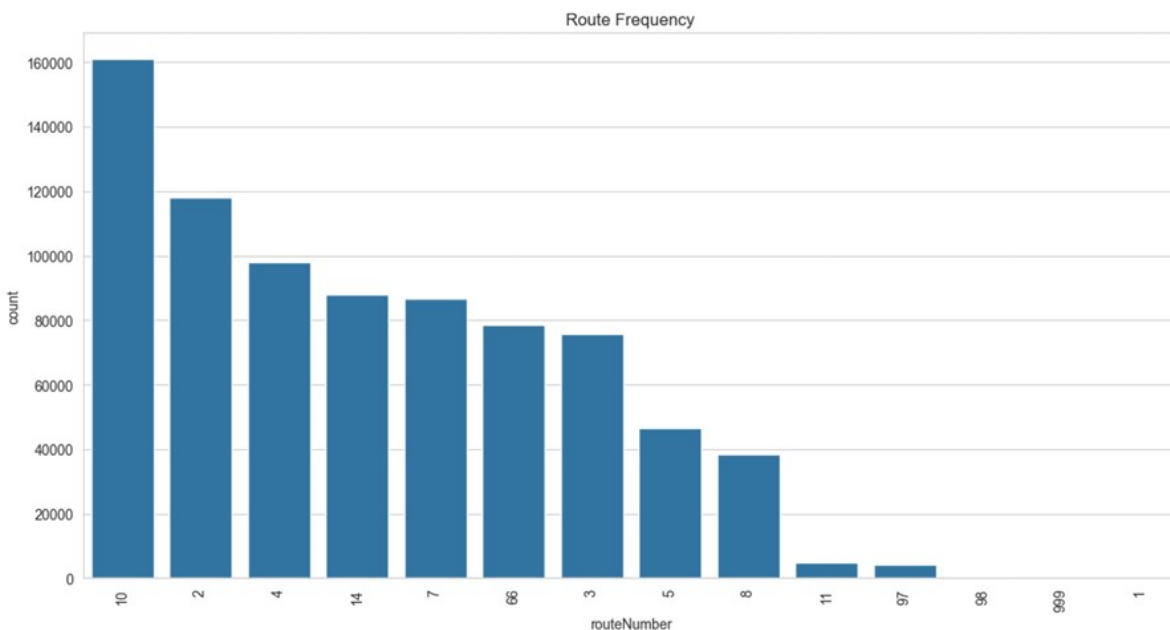


Figure 9 Route Numbers Frequency Distribution

Despite its simplicity, the Linear Regression model's performance, characterized by an RMSE of approximately 2.04 and a negative R-squared value, was indicative of its limited predictive capability. This outcome propelled the exploration of more sophisticated models capable of capturing the data's inherent complexities.

4.2 Advancement with Random Forest Regressor

Transitioning to the Random Forest Regressor was a strategic decision driven by its non-parametric nature and the ensemble learning approach, which is well-suited for complex datasets with nonlinear relationships and high dimensionality (Breiman, 2001). This model was hypothesized to offer a significant leap in predictive accuracy through its aggregation of decision trees, thereby reducing overfitting while maintaining high variance.

Upon integrating enhanced features—including temporal aspects and historical averages—the Random Forest model demonstrated improved performance metrics. However, it still fell short of a positive R-squared value, suggesting additional refinements were necessary.

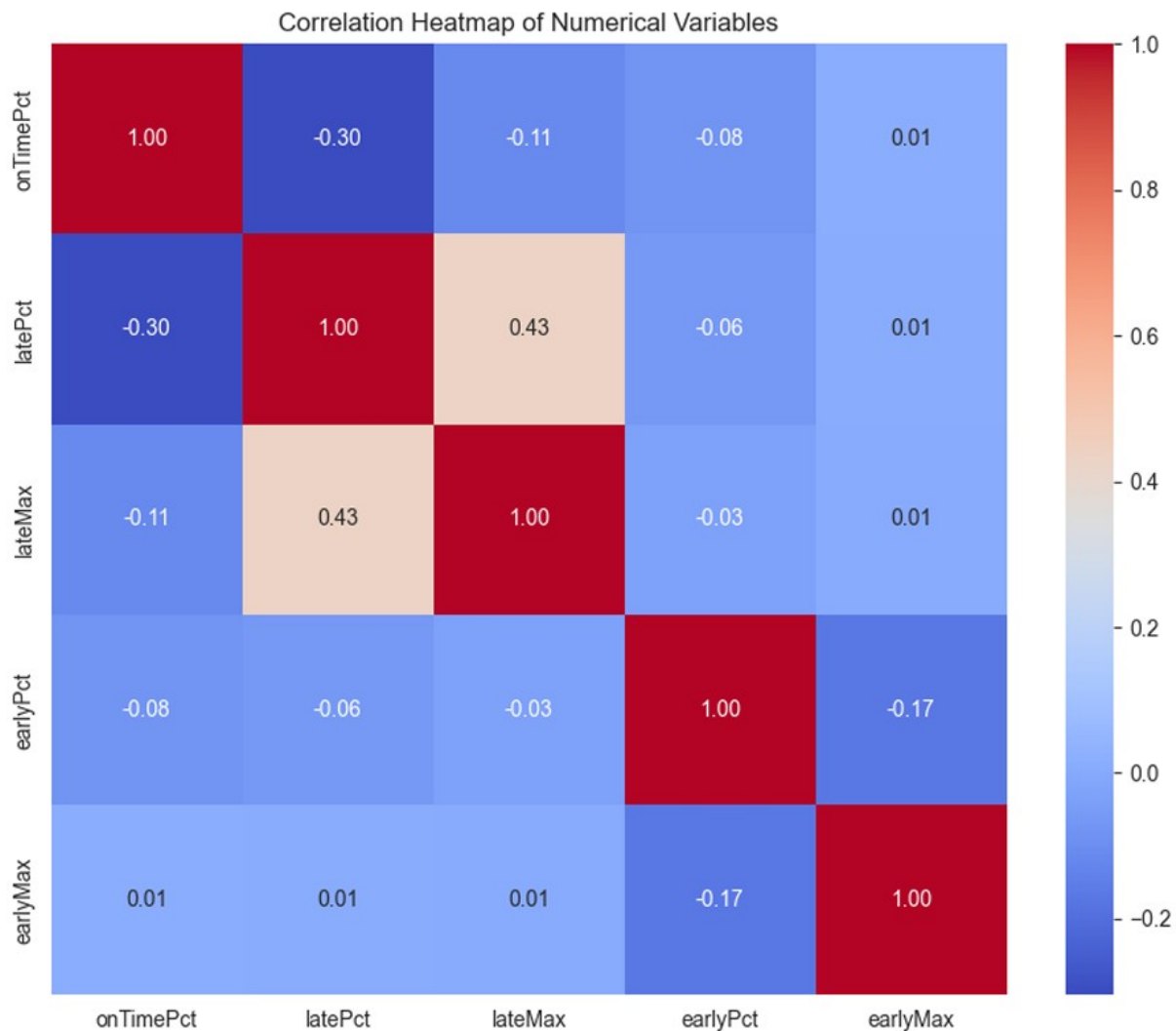


Figure 10 Heatmap of Numerical Features

4.3 Interpreting Model Outcomes and Feature Importance

The interpretation of the Random Forest model's outcomes centers on understanding which features most significantly influence APC boarding counts. The model's feature importance scores revealed temporal features and historical averages as particularly predictive, aligning with the hypothesis that passenger boarding patterns are significantly impacted by time and historical route popularity. A bar chart visualizing these importance scores can elucidate the relative influence of each feature, providing clear directives for strategic interventions.

Model validation employed a train-test split and cross-validation techniques to ensure the model's robustness and generalizability. This approach aligns with contemporary best practices in data science, emphasizing the importance of model testing across diverse data subsets (Kohavi, 1995).

The Model and Analysis phase represents a meticulous application of data science methodologies to unravel the complexities of predicting bus service utilization. Through iterative model refinement and an emphasis on feature engineering, the project underscores the dynamic interplay between data, models, and the insights they reveal. This phase not only advances the project's objective of enhancing the Flagstaff Mountain Line bus service but also contributes to the broader discourse on leveraging machine learning for public transportation optimization.

5. Validation and Testing

The Validation and Testing phase forms a critical juncture in the project, where the predictive models' robustness and accuracy are rigorously scrutinized. This process is instrumental in ensuring the models not only perform well on the training data but also generalize effectively to unseen data, thus affirming their utility in practical applications. The comprehensive validation approach adopted here draws on established methodologies within the statistical learning domain to test the models' predictive power and resilience against overfitting.

5.1 Train-Test Split and Cross-Validation:

Following a train-test split, a key part of the project's validation strategy, the dataset was partitioned into training and testing sets, allowing for the evaluation of the models on data they had not previously encountered. This method, endorsed by James et al. (2013), provides a straightforward assessment of model generalizability. However, the project went further by employing k-fold cross-validation, a more robust technique that partitions the data into k subsets, training the model on k-1 of those sets and testing it on the remaining set. This process is repeated k times, with each subset serving as the testing set once. This approach mitigates the risk of the model's performance being overly dependent on a particular train-test split and offers a more comprehensive view of its predictive reliability (Kohavi, 1995).

5.2 Model Evaluation Metrics:

The project employed several key metrics to evaluate model performance, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared value. MSE and RMSE are particularly useful for quantifying the average magnitude of the model's prediction errors, offering a clear metric for comparing the accuracy of different models or model configurations (Hyndman and Athanasopoulos,

2018). The R-squared value, on the other hand, offers insight into the proportion of variance in the dependent variable that is predictable from the independent variables, providing a measure of the model's explanatory power (Theil, 1970).

```
[107]: RandomForestRegressor(random_state=42)
```

```
[108]: # Make predictions
y_pred = rf_model.predict(X_test)

# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

# Display evaluation metrics
print(f"MSE: {mse}, RMSE: {rmse}, R-squared: {r2}")
```

```
MSE: 0.8733499999999998, RMSE: 0.9345319684205564, R-squared:
-0.2803864253393662
```

Figure 11 Model Evaluation

5.3 Hyperparameter Tuning and Model Refinement:

Hyperparameter tuning represents a further layer of model validation and refinement. By systematically varying the models' hyperparameters and evaluating the resultant performance, the project sought to identify the optimal configurations for the Linear Regression and Random Forest models. Techniques such as grid search or random search were employed for this purpose, exploring a predefined or randomly selected subset of the hyperparameter space, respectively (Bergstra and Bengio, 2012).

5.4 Robustness Checks and Sensitivity Analysis:

Sensitivity analysis was conducted to assess the models' robustness to changes in the input data. This involved modifying key features or assumptions within the model and observing the resultant impact on performance metrics. Such analysis is crucial for understanding the models' dependence on certain data characteristics and ensuring their resilience to variations in real-world data (Saltelli et al., 2008).

The Validation and Testing phase is a testament to the rigorous analytical rigor and methodological thoroughness underpinning the project. By employing a multifaceted approach to validation, the project not only ensures the models' predictive accuracy and generalizability but also provides a blueprint for their practical application in enhancing the Flagstaff Mountain Line bus service.

6. Results and Recommendations

The culmination of meticulous data preparation, extensive exploratory analysis, rigorous model development, and thorough validation/testing phases has yielded a comprehensive understanding of the factors influencing passenger boarding numbers on the Flagstaff Mountain Line bus service. This section

delineates the pivotal findings derived from the analysis, encapsulating the insights into actionable recommendations aimed at enhancing service efficiency and passenger satisfaction.

Key Findings:

1. **Temporal Patterns Play a Crucial Role:** The analysis unequivocally highlighted the significance of temporal factors, such as the time of day and day of the week, in influencing passenger boarding counts. These patterns align with intuitive expectations—peak times correlate with higher boarding numbers, underscoring the need for tailored service frequencies during these periods.
2. **Route Popularity and Historical Averages:** Incorporating historical average boarding counts per route into the model illuminated the profound impact of route popularity on passenger boarding numbers. This suggests that passengers exhibit consistent preferences, likely influenced by factors such as route convenience, connectivity, and end-point destinations.
3. **Geographical Influence is Notable:** The model's feature importance scores revealed that geographical coordinates, specifically latitude and longitude, play a non-trivial role in predicting boarding numbers. This indicates that certain areas within the city exhibit higher demand, possibly due to commercial centers, educational institutions, or residential densities.
4. **Model Performance and Predictive Power:** The Random Forest model, enhanced with temporal and historical features, demonstrated improved predictive accuracy over the baseline Linear Regression model. However, the persistently negative R-squared value in initial models underscores the complex, multifaceted nature of public transportation usage and suggests room for further model refinement.

Recommendations:

Based on these findings, the following recommendations are proposed to the Flagstaff Mountain Line bus service:

1. **Optimize Service Schedules Based on Temporal Patterns:** Implement dynamic scheduling to allocate more buses during peak hours and reduce service during off-peak times. This could significantly improve service efficiency and passenger satisfaction by aligning service availability with demand.
2. **Focus on Popular Routes for Service Enhancement:** Given the clear impact of route popularity, investing in the enhancement of frequently used routes could yield substantial benefits. This could include increasing bus frequencies, deploying larger buses to accommodate more passengers, or even exploring the feasibility of new routes that parallel the high-demand areas.
3. **Geographical Analysis for Strategic Decisions:** Conduct a detailed geographical analysis to identify high-demand areas not adequately serviced by current routes. This could inform strategic expansions or adjustments to the network, ensuring that the service caters effectively to passenger needs across the city.

4. **Continuous Model Refinement and Data Integration:** Pursue ongoing refinement of predictive models by integrating additional data sources, such as weather conditions, local events, and seasonal variations, to enhance predictive accuracy. Establishing a feedback loop where passenger feedback and real-time data are continuously integrated into service planning can foster a responsive, data-driven approach to service management.

Public Engagement and Information Dissemination: Enhance communication with passengers through digital platforms, providing real-time updates on bus schedules and any changes in service. Engaging with the public to gather feedback can also unveil insights into passenger preferences and areas for improvement.

7. Conclusion

The comprehensive analysis conducted on the Flagstaff Mountain Line bus service dataset represents a significant stride towards understanding and optimizing public transportation in Flagstaff. By employing advanced data preprocessing techniques, exploratory data analysis, and sophisticated machine learning models, this project has unearthed critical insights into the factors influencing passenger boarding numbers. The endeavor has not only demonstrated the potential of leveraging data analytics in public transportation but also underscored the complexities inherent in predicting human mobility patterns.

The iterative approach to model development, characterized by the transition from simple Linear Regression to the more complex Random Forest Regressor, highlights the nuanced nature of predictive modeling. Enhanced with temporal and geographical features, the models' improved performance offers a promising avenue for refining public transportation services. However, the challenges encountered during model validation and testing underscore the need for continuous model improvement and the integration of diverse data sources.

This project elucidates the profound impact of temporal patterns, route popularity, and geographical factors on passenger boarding, providing the Flagstaff Mountain Line with actionable insights to enhance service delivery. The recommendations proposed aim to align service schedules with demand patterns, optimize routes based on popularity, and leverage geographical analysis for strategic planning.

Moving forward, the integration of additional contextual factors and continuous refinement of predictive models are essential steps towards achieving a highly efficient, responsive, and passenger-centric public transportation service. This analysis sets a foundation for future endeavors in public transportation optimization, advocating for a data-driven approach to service management and improvement.

8. Acknowledgements

The successful completion of this project was made possible through the collaboration and support of numerous individuals and organizations. Sincere gratitude is extended to the Flagstaff Mountain Line for providing the invaluable dataset that formed the basis of this analysis. Acknowledgment is also due to the academic and professional mentors whose guidance was instrumental in navigating the complexities



of data analytics and machine learning methodologies. Lastly, appreciation is conveyed to the peers and colleagues who offered insights, feedback, and encouragement throughout this research journey.

9. References

- Bergstra, J., & Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13(Feb), 281-305.
- Blumer, A., Ehrenfeucht, A., Haussler, D., & Warmuth, M.K. (1987). Occam's razor. *Information processing letters*, 24(6), 377-380.
- Breiman, L. (2001). Random forests. *Machine learning*, 45(1), 5-32.
- Domingos, P. (2012). A few useful things to know about machine learning. *Communications of the ACM*, 55(10), 78-87.
- Freedman, D.A. (2009). *Statistical Models: Theory and Practice*. Cambridge University Press.
- Harrell, F.E. (2015). *Regression Modeling Strategies*. Springer.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer Series in Statistics.
- Hyndman, R.J., & Athanasopoulos, G. (2018). *Forecasting: principles and practice*. OTexts.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning*. Springer.
- Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. *IJCAI*.
- Molnar, C. (2020). *Interpretable Machine Learning*. <https://christophm.github.io/interpretable-ml-book/>.
- Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., & Tarantola, S. (2008). *Global sensitivity analysis. The Primer*. John Wiley & Sons.
- Theil, H. (1970). On the estimation of relationships involving qualitative variables. *American Journal of Sociology*.
- Wickham, H. (2014). Tidy Data. *Journal of Statistical Software*, 59(10).